GAMAL TAWAF

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PROFESSIONAL SUMMARY

Senior Software Engineer and Google Cloud Certified **Professional Cloud Architect** with 12+ years building mission-critical, high-reliability backend systems, global MasterCard issuance at Binance, financial-services platforms at Fiserv, consumer platforms serving 10M+ users, and a GCP platform run at 99.9% uptime. Deep expertise in distributed systems, observability, and GCP cloud architecture, working primarily in Python today after a decade across PHP, Ruby, and Java. Proven technical lead who has hired and grown engineering teams and repeatedly turned months-long integration work into weeks.

TECHNICAL SKILLS

Backend: Python, FastAPI, Flask, Django, Java, Spring Boot, PHP, Laravel, REST API design

Cloud & Architecture: Google Cloud Platform (Cloud Run, Pub/Sub, BigQuery, Vertex AI), Kubernetes, Docker, Terraform, microservices, distributed systems, high-availability design

Data & Real-Time: PostgreSQL, MySQL, Redis (hot-state / caching), BigQuery, Google Cloud Pub/Sub, real-time event pipelines

Observability & Delivery: OpenTelemetry distributed tracing, monitoring & alerting, GitHub Actions, GitLab CI/CD, pytest, JUnit, test automation

AI Engineering: Vertex AI, Gemini, LangChain, RAG systems, AI data pipelines

EXPERIENCE

Senior Software Engineer, Truss Payments

Austin, Texas / Remote | Aug 2026 – Oct 2026

- Cut local setup for new engineers to under an hour by introducing a Docker-based development environment.
- Proposed a GCP infrastructure plan estimated to cut the cloud bill by 50%.
- Audited the cloud infrastructure for compliance and wrote the path to close the gaps.
- Took on-call and fixed customer payment issues in production.
- Set the team rules for agent-written code so new agent work follows existing conventions.
- Reworked unfinished payment and account screens so the OTP step shows a loading state and the number entered, actions a user cannot take stay hidden, blocked steps notify the person who can clear them, and failures say what went wrong.

Senior Software Engineer, Fiserv (via Ascendion)

Toronto, ON | Jan 2026 – Aug 2026

- Engineer on a core financial-services platform (Python, Flask, MySQL, Docker) at one of the world's largest fintech providers, serving as Feature Lead for multiple core platform features and owning cross-team delivery.
- Introduced OpenTelemetry distributed tracing with traceparent propagation across 24 microservices, cutting production issue diagnosis time by 80%, from several hours of cross-service investigation to under an hour.
- Re-architected vendor integration across 5 microservices into a configuration-driven design, cutting new-vendor onboarding from ~2 months of cross-service code changes to ~2 weeks of configuration.
- Led a new vendor integration initiative end-to-end, delivering a customer-facing capability designed to drive client retention and product adoption.
- Rebuilt developer onboarding, cutting new-hire environment setup from up to a week to under one day.
- Resolved on-call production incidents on a platform where reliability directly affects financial transactions.

Solution Architect / Engineering Team Lead, Inkeros

Etobicoke, ON | Jan 2025 – Jan 2026

- Recruited, led, and mentored the full engineering team (junior through senior) while owning architecture for the company's core product.
- Architected a mission-critical enterprise portal (Python/Django on GCP) automating sales-reporting workflows for 250+ staff, eliminating ~2 hours/day of manual submission per sales rep and hours of manager spreadsheet work, reducing the entire workflow to a 5–10 minute approval. Designed SaaS-ready for multi-tenant expansion.
- Built a fully automated CI/CD pipeline (Docker, GitHub Actions, Artifact Registry, Cloud Run) delivering one-click deployments at a 99.8% success rate.

- Designed and piloted an AI compliance system analyzing call-center audio with Gemini and Vertex AI, orchestrated via GCP Workflows into BigQuery, architected to scale to a 100-agent, 6-day/week calling operation for fraud prevention and compliance insight.
- Built generative-AI data pipelines (Pandas, GeoPandas, SERP/Maps APIs) that quadrupled lead-profiling throughput, and prototyped a LangChain + Vertex AI RAG system.
- Ran the platform at 99.9% uptime on GCP autoscaling infrastructure and cut mean time to resolution by 60% through improved monitoring and runbooks.

Senior Software Engineer, PERCS

Remote | Apr 2024 – Jan 2025 (startup, role eliminated in layoffs; company ceased operations in 2025)

- Scaled full-stack platform features through rapid user growth and integrated product analytics to guide roadmap decisions.
- Designed and launched Web3 games on blockchain-based social protocols (Warpcast Frames) during the protocol's early growth stage, driving strong user engagement and raising the company's visibility in the ecosystem.

Senior Software Engineer, University of Toronto (via Tundra)

Toronto, ON | Jan 2024 – Apr 2024 (3-month contract)

- Delivered a fixed-scope modernization: Laravel v8 to v10 upgrade, unit testing introduced to 90% coverage, and local development modernized with Sail.

Senior Software Engineer, Binance, Card & Pay

Remote | Jul 2022 – Jul 2023 (role impacted by 2023 global workforce reduction)

- Integrated 3 global payment partners' REST APIs to enable MasterCard issuance across Europe, the Middle East, and South America at one of the world's largest crypto exchanges.
- Migrated the card-ordering service from Laravel to Java/Spring Boot, cutting new payment-partner onboarding from ~3 months to ~1 week and improving performance and scalability of a revenue-critical flow.
- Built the new Spring Boot services with unit testing from day one (vs. an untested legacy codebase), and trained team members in Java and Spring Boot during the migration.

Senior Web Engineer, VerticalScope Inc.

Remote | Jul 2021 – Jun 2022 (recruited into Binance)

- Designed and shipped features for community platforms serving 10M+ users, and launched the platform's first marketplace product, a zero-to-one build converting ad-hoc forum-post listings into a structured commerce experience and new revenue stream.

Team Lead / Senior Software Developer, Afilias Canada

Toronto, ON | Feb 2016 – Jul 2021

- Led the in-house migration of a business-critical Ruby on Rails application off a third-party vendor, upgrading Rails v3 to v5 and migrating MySQL to PostgreSQL in 6 months, delivering $500K in annual savings and full operational control.
- Took ownership of 3 previously outsourced systems and built the internal team's capability to run them, establishing lasting business confidence in in-house engineering.
- Hired, trained, and led a team of 5–7 developers over 5+ years, growing engineers from junior to senior levels.

Earlier Roles (2013–2016)

Carlton Group | Golf Canada | PAVR Labs

LANGUAGES

Arabic, native fluency | English, fluent

CERTIFICATIONS

Google Cloud Professional Cloud Architect, 2025–2027
Hugging Face AI Agents Course, Jul 2026

EDUCATION

Bachelor of Computer Science, York University, Toronto, ON
Computer Programming Diploma, Seneca College, Toronto, ON